



# JABchem



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# Past Papers Advanced Higher Chemistry

# 2019 Marking Scheme

| Grade Awarded | Mark Required |        | % candidates achieving grade |
|---------------|---------------|--------|------------------------------|
|               | (/130)        | %      |                              |
| A             | 90+           | 69.2%  | 33.8%                        |
| B             | 77+           | 59.2%  | 26.3%                        |
| C             | 64+           | 49.2%  | 22.2%                        |
| D             | 57+           | 43.8%  | 7.3%                         |
| No award      | <57           | <43.8% | 10.4%                        |

| Section:      | Multiple Choice | Extended Answer | Assignment |
|---------------|-----------------|-----------------|------------|
| Average Mark: | 21.0 /30        | 40.4 /70        | 19.2 /30   |

| 2019 Adv Higher Chemistry Marking Scheme |                                                                                                                 |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----|------------------------------|------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---|------------------|--|---|-----------|---|------------|--|---|------|---|--------|--|---|--------------------------|--|--|
| MC Qu                                    | Answer                                                                                                          | % Pupils Correct                                                                                | Reasoning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 1                                        | B                                                                                                               | 76                                                                                              | <input checked="" type="checkbox"/> A adsorption spectra formed by adsorption of energy as electrons are promoted<br><input checked="" type="checkbox"/> B emission spectra are formed from the release of energy as electrons drop down<br><input checked="" type="checkbox"/> C 310nm is a wavelength in the ultraviolet not visible region of EM Spectrum<br><input checked="" type="checkbox"/> D 310nm is a wavelength in the ultraviolet not visible region of EM Spectrum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 2                                        | D                                                                                                               | 85                                                                                              | In gravimetric analysis, heating Na <sub>2</sub> CO <sub>3</sub> .10H <sub>2</sub> O will release the 10 water molecules trapped in the structure so that just Na <sub>2</sub> CO <sub>3</sub> remains. The substance is heated, cooled in a desiccator and its mass measured. The process is repeated until the mass is constant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 3                                        | C                                                                                                               | 37                                                                                              | <input checked="" type="checkbox"/> A adding sodium nitrate to magnesium ions does not produce a magnesium precipitate<br><input checked="" type="checkbox"/> B adding silver(I) nitrate to magnesium ions does not produce a magnesium precipitate<br><input checked="" type="checkbox"/> C sodium carbonate forms a magnesium carbonate precipitate with magnesium ion solution<br><input checked="" type="checkbox"/> D silver(I) carbonate is insoluble so carbonate ions cannot precipitate with magnesium ions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 4                                        | A                                                                                                               | 90                                                                                              | <input checked="" type="checkbox"/> A Hund's Rule: electrons occupy degenerate orbitals singly with parallel spins before pairing<br><input checked="" type="checkbox"/> B Pauli Exclusion Principle: no two electrons in an atom can have the same set of four quantum numbers<br><input checked="" type="checkbox"/> C Aufbau Principle: electrons occupy orbitals in order of increasing energy<br><input checked="" type="checkbox"/> D the energy of an electron in an atom is quantised into specific energy levels/electron shells                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 5                                        | D                                                                                                               | 75                                                                                              | <input checked="" type="checkbox"/> A BeCl <sub>2</sub> is a linear molecule with 180° angles<br><input checked="" type="checkbox"/> B BCl <sub>3</sub> is a trigonal planar molecule with 120° angles between bonds<br><input checked="" type="checkbox"/> C CCl <sub>4</sub> is a tetrahedral molecule with 109.5° angles between bonds<br><input checked="" type="checkbox"/> D PCl <sub>5</sub> is a trigonal pyramidal molecule with 90° (and 120°) angles between bonds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 6                                        | D                                                                                                               | 51                                                                                              | <table><tr><th>Species</th><th>Fe</th><th>Fe<sup>2+</sup></th><th>Fe<sup>3+</sup></th></tr><tr><td>Electronic configuration</td><td>1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup> 2p<sup>6</sup> 3d<sup>6</sup> 4s<sup>2</sup></td><td>1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup> 2p<sup>6</sup> 3d<sup>6</sup></td><td>1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup> 2p<sup>6</sup> 3d<sup>5</sup></td></tr></table> <input checked="" type="checkbox"/> A Fe <sup>2+</sup> and Fe <sup>3+</sup> both have three occupied energy levels (n=1, n=2 and n=3)<br><input checked="" type="checkbox"/> B Fe <sup>2+</sup> has four unpaired electrons in 3d and Fe <sup>3+</sup> has five unpaired electrons in 3d<br><input checked="" type="checkbox"/> C Fe <sup>2+</sup> is higher in ECS than Fe <sup>3+</sup> . Fe <sup>3+</sup> is better oxidising agent than reducing agent<br><input checked="" type="checkbox"/> D Fe <sup>3+</sup> more stable because Fe <sup>3+</sup> had 5 unpaired electrons in half-filled d-subshell | Species                       | Fe | Fe <sup>2+</sup>             | Fe <sup>3+</sup> | Electronic configuration      | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>6</sup> 4s <sup>2</sup> | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>6</sup> | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>5</sup> |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| Species                                  | Fe                                                                                                              | Fe <sup>2+</sup>                                                                                | Fe <sup>3+</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| Electronic configuration                 | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>6</sup> 4s <sup>2</sup> | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>6</sup> | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>2</sup> 2p <sup>6</sup> 3d <sup>5</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 7                                        | B                                                                                                               | 71                                                                                              | <input checked="" type="checkbox"/> A Cr <sub>2</sub> O <sub>7</sub> <sup>2-</sup> ions have Cr with an oxidation state of +6<br><input checked="" type="checkbox"/> B MnO <sub>4</sub> <sup>-</sup> ions have Mn with an oxidation state of +7<br><input checked="" type="checkbox"/> C VO <sup>2+</sup> ions have V with an oxidation state of +4<br><input checked="" type="checkbox"/> D Sn <sup>4+</sup> ions have Sn with an oxidation state of +4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 8                                        | D                                                                                                               | 60                                                                                              | <input checked="" type="checkbox"/> A Co-ordination number of the copper 4 as there are four bonds from ligand to Cu<br><input checked="" type="checkbox"/> B Co-ordination number of the copper 4 as there are four bonds from ligand to Cu<br><input checked="" type="checkbox"/> C The ligand shares four pairs of electrons with the one metal ion = tetradentate<br><input checked="" type="checkbox"/> D Four bonds between ligand and central metal ion: co-ordination number equals 4 and is tetradentate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 9                                        | A                                                                                                               | 72                                                                                              | <input checked="" type="checkbox"/> A Decrease in temperature favours the forward exothermic reaction. This gives more products and increasing the numerator on top of the equilibrium constant calculation which increases the value of equilibrium constant.<br><input checked="" type="checkbox"/> B Increase in temperature favours the reverse endothermic reaction. Lowers K value<br><input checked="" type="checkbox"/> C Changes in pressure do not alter the equilibrium constant<br><input checked="" type="checkbox"/> D Changes in pressure do not alter the equilibrium constant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 10                                       | C                                                                                                               | 72                                                                                              | <input checked="" type="checkbox"/> A The reaction is feasible when the value of ΔG is below zero<br><input checked="" type="checkbox"/> B The reaction is only feasible when the value of ΔG is below zero (above 300K)<br><input checked="" type="checkbox"/> C The value of ΔG is negative above 300K so reaction is feasible above 300K<br><input checked="" type="checkbox"/> D The value of ΔG is positive below 300K so reaction is <u>not</u> feasible below 300K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
| 11                                       | C                                                                                                               | 89                                                                                              | <table><tr><td>ΔH°</td><td>=</td><td>ΣΔH<sub>f</sub>°(products)</td><td>-</td><td>ΣΔH<sub>f</sub>°(reactants)</td></tr><tr><td></td><td>=</td><td>(2x0) + (3x-242)</td><td>-</td><td>(1x-822) + (3x0)</td></tr><tr><td></td><td>=</td><td>(0 - 726)</td><td>-</td><td>(-822 + 0)</td></tr><tr><td></td><td>=</td><td>-726</td><td>-</td><td>(-822)</td></tr><tr><td></td><td>=</td><td>+96 kJ mol<sup>-1</sup></td><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ΔH°                           | =  | ΣΔH <sub>f</sub> °(products) | -                | ΣΔH <sub>f</sub> °(reactants) |                                                                                                                 | =                                                                                               | (2x0) + (3x-242)                                                                                | - | (1x-822) + (3x0) |  | = | (0 - 726) | - | (-822 + 0) |  | = | -726 | - | (-822) |  | = | +96 kJ mol <sup>-1</sup> |  |  |
| ΔH°                                      | =                                                                                                               | ΣΔH <sub>f</sub> °(products)                                                                    | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ΣΔH <sub>f</sub> °(reactants) |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
|                                          | =                                                                                                               | (2x0) + (3x-242)                                                                                | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (1x-822) + (3x0)              |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
|                                          | =                                                                                                               | (0 - 726)                                                                                       | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (-822 + 0)                    |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
|                                          | =                                                                                                               | -726                                                                                            | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (-822)                        |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |
|                                          | =                                                                                                               | +96 kJ mol <sup>-1</sup>                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                               |    |                              |                  |                               |                                                                                                                 |                                                                                                 |                                                                                                 |   |                  |  |   |           |   |            |  |   |      |   |        |  |   |                          |  |  |

| 12                   | B                                                                                                               | 39             | <div><input checked="" type="checkbox"/>A Steam condensing into water gives off heat <math>\therefore \Delta H</math> is negative</div> <div><input checked="" type="checkbox"/>B <math>\Delta H</math> is negative as heat is given off. <math>\Delta S</math> is negative as molecules are more ordered</div> <div><input checked="" type="checkbox"/>C Steam condensing into water gives off heat <math>\therefore \Delta H</math> is negative</div> <div><input checked="" type="checkbox"/>D Molecules are closer together and more ordered in water <math>\therefore \Delta S</math> is negative</div>                                                                                                                                   |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
|----------------------|-----------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------|----------------------|---------|-----|----------------|-----------|---------|----------------|---|---|---|---|---|
| 13                   | A                                                                                                               | 86             | <table><tr><th>Experiment</th><th>Change</th><th>Effect on Rate</th><th>Order of reactant</th></tr><tr><td>1+2</td><td><math>[X] \times 2</math></td><td><math>\times 2</math></td><td><math>[X]^1</math></td></tr><tr><td>1+2</td><td><math>[Y] \times 2</math></td><td>No effect</td><td><math>[Y]^0</math></td></tr></table> $\text{Rate} = k [X]^1 [Y]^0 = k[X]$                                                                                                                                                                                                                                                                                                                                                                           | Experiment | Change                                                                                   | Effect on Rate | Order of reactant                                                                                               | 1+2    | $[X] \times 2$                                                                        | $\times 2$           | $[X]^1$ | 1+2 | $[Y] \times 2$ | No effect | $[Y]^0$ |                |   |   |   |   |   |
| Experiment           | Change                                                                                                          | Effect on Rate | Order of reactant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 1+2                  | $[X] \times 2$                                                                                                  | $\times 2$     | $[X]^1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 1+2                  | $[Y] \times 2$                                                                                                  | No effect      | $[Y]^0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 14                   | B                                                                                                               | 75             | <div><math display="block">\begin{aligned} \text{rate} &amp;= k [A]^1 [B]^1 \\ \therefore k &amp;= \frac{\text{rate}}{[A]^1 [B]^1} \\ &amp;= \frac{\text{mol l}^{-1} \text{ s}^{-1}}{\text{mol l}^{-1} \times \text{mol l}^{-1}} \\ &amp;= \text{l mol}^{-1} \text{ s}^{-1} \end{aligned}</math></div> <div><math display="block">\begin{aligned} \text{rate} &amp;= k [C]^2 \\ \therefore k &amp;= \frac{\text{rate}}{[C]^2} \\ &amp;= \frac{\text{mol l}^{-1} \text{ s}^{-1}}{\text{mol}^2 \text{ l}^{-2}} \\ &amp;= \text{l mol}^{-1} \text{ s}^{-1} \end{aligned}</math></div>                                                                                                                                                             |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 15                   | C                                                                                                               | 78             | <table><tr><th>Bonds</th><th>2x C=C</th><th>2x C-C</th><th>1x C≡N</th><th>2x C-O</th><th>3x C-H</th></tr><tr><td>Sigma <math>\sigma</math> bonds</td><td>2</td><td>2</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Pi <math>\pi</math> bonds</td><td>2</td><td>0</td><td>2</td><td>0</td><td>0</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                              | Bonds      | 2x C=C                                                                                   | 2x C-C         | 1x C≡N                                                                                                          | 2x C-O | 3x C-H                                                                                | Sigma $\sigma$ bonds | 2       | 2   | 1              | 2         | 3       | Pi $\pi$ bonds | 2 | 0 | 2 | 0 | 0 |
| Bonds                | 2x C=C                                                                                                          | 2x C-C         | 1x C≡N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2x C-O     | 3x C-H                                                                                   |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| Sigma $\sigma$ bonds | 2                                                                                                               | 2              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2          | 3                                                                                        |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| Pi $\pi$ bonds       | 2                                                                                                               | 0              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0          | 0                                                                                        |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 16                   | A                                                                                                               | 89             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 17                   | C                                                                                                               | 49             | <div><input checked="" type="checkbox"/>A The right C in C=C double bond has 2 -H groups <math>\therefore</math> no geometric isomerism</div> <div><input checked="" type="checkbox"/>B The right C in C=C double bond has 2 -CH<sub>3</sub> groups <math>\therefore</math> no geometric isomerism</div> <div><input checked="" type="checkbox"/>C Both ends of C=C double bond have different groups <math>\therefore</math> has geometric isomerism</div> <div><input checked="" type="checkbox"/>D The right C in C=C double bond has 2 -H groups <math>\therefore</math> no geometric isomerism</div>                                                                                                                                      |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 18                   | C                                                                                                               | 80             | <div><div><b>Structure:</b> 3,4-dimethylhex-3-ene<br/>Both ethyl groups are on same side of C=C double bond so this is the CIS geometric isomer.<br/><b>Name:</b> cis-3,4-dimethylhex-3-ene</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 19                   | D                                                                                                               | 59             | <div><input checked="" type="checkbox"/>A Trimethylamine is a tertiary amine so no hydrogen bonding <math>\therefore</math> lower boiling point</div> <div><input checked="" type="checkbox"/>B Trimethylamine is a tertiary amine so no hydrogen bonding <math>\therefore</math> lower boiling point</div> <div><input checked="" type="checkbox"/>C Trimethylamine has no H-N bonds so lacks any hydrogen bonding <math>\therefore</math> less soluble</div> <div><input checked="" type="checkbox"/>D Trimethylamine has no hydrogen bonding as it lacks a H-N bond. This means trimethylamine molecules are further apart and this lowers the boiling point. The lack of N-H bonds lowers the solubility of trimethylamine in water.</div> |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 20                   | B                                                                                                               | 78             | <p>Lithium Aluminium Hydride is a reducing agent for:</p> <div><div>carboxylic acid</div><div>ketone</div><div><math>\longrightarrow</math></div><div>aldehyde</div><div>secondary alcohol</div><div><math>\longrightarrow</math></div><div>primary alcohol</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 21                   | C                                                                                                               | 56             | <table><tr><td>Step 1</td><td>Electrophilic substitution: -NO<sub>2</sub> group is substituted onto the benzene ring.</td></tr><tr><td>Step 2</td><td>Reduction: Decrease in the oxygen:hydrogen ratio as -NO<sub>2</sub> group is reduced to -NH<sub>2</sub> group</td></tr><tr><td>Step 3</td><td>Condensation: two molecules join together and small molecule removes where they join.</td></tr></table>                                                                                                                                                                                                                                                                                                                                    | Step 1     | Electrophilic substitution: -NO <sub>2</sub> group is substituted onto the benzene ring. | Step 2         | Reduction: Decrease in the oxygen:hydrogen ratio as -NO <sub>2</sub> group is reduced to -NH <sub>2</sub> group | Step 3 | Condensation: two molecules join together and small molecule removes where they join. |                      |         |     |                |           |         |                |   |   |   |   |   |
| Step 1               | Electrophilic substitution: -NO <sub>2</sub> group is substituted onto the benzene ring.                        |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| Step 2               | Reduction: Decrease in the oxygen:hydrogen ratio as -NO <sub>2</sub> group is reduced to -NH <sub>2</sub> group |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| Step 3               | Condensation: two molecules join together and small molecule removes where they join.                           |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 22                   | B                                                                                                               | 63             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |
| 23                   | B                                                                                                               | 64             | <p>The heaviest <math>m/z</math> peak on a mass spectrum is the mass of the original compound.</p> <p>gfm C<sub>3</sub>H<sub>6</sub>O = (3×12)+(6×1)+(1×16) = 36+6+16 = 58g</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                                                                                          |                |                                                                                                                 |        |                                                                                       |                      |         |     |                |           |         |                |   |   |   |   |   |

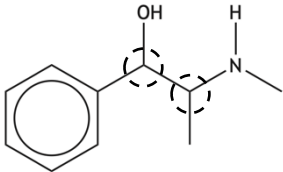
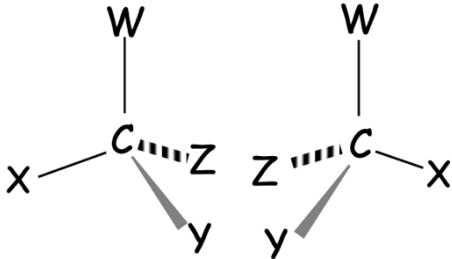
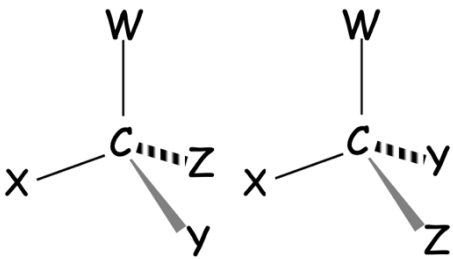
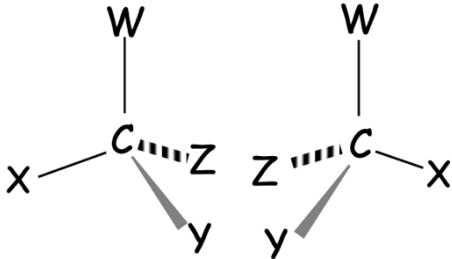
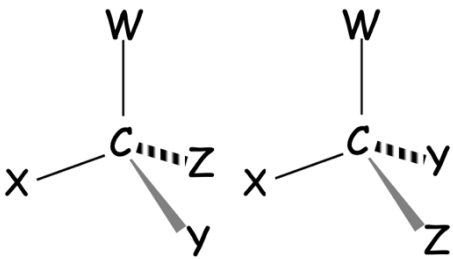
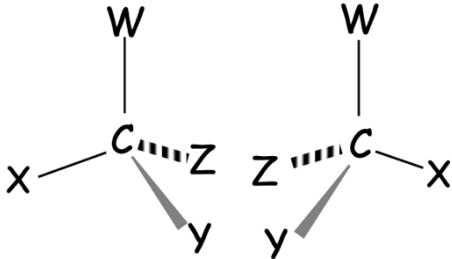
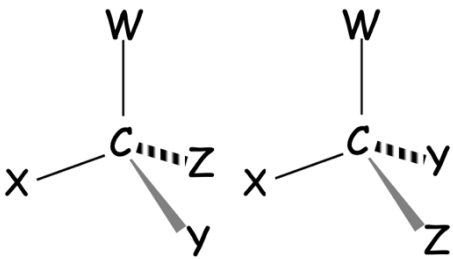
|                                                  |                                 |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
|--------------------------------------------------|---------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|---|---|------|-----|------|-----------------------------------|-----------------------------|----------------------------|------------------------------|--------------------------------------------------|---------------------------------|--------------------------------|-------------------------------|-----------------------|----|----|---|
| 24                                               | A                               | 90                             | <table><tr><td>Element</td><td>C</td><td>H</td><td>O</td></tr><tr><td>%</td><td>80.0</td><td>9.3</td><td>10.7</td></tr><tr><td>No. of moles<br/>(divide % by gfm)</td><td><math>\frac{80.0}{12}</math><br/>= 6.67</td><td><math>\frac{9.3}{1}</math><br/>= 9.300</td><td><math>\frac{10.7}{16}</math><br/>= 0.669</td></tr><tr><td>Mole ratio<br/>(divide through by smallest value)</td><td><math>\frac{6.667}{0.669}</math><br/>= 9.96</td><td><math>\frac{9.300}{0.669}</math><br/>13.90</td><td><math>\frac{0.669}{0.669}</math><br/>1.00</td></tr><tr><td>Round to Whole Number</td><td>10</td><td>14</td><td>1</td></tr></table> | Element | C | H | O | % | 80.0 | 9.3 | 10.7 | No. of moles<br>(divide % by gfm) | $\frac{80.0}{12}$<br>= 6.67 | $\frac{9.3}{1}$<br>= 9.300 | $\frac{10.7}{16}$<br>= 0.669 | Mole ratio<br>(divide through by smallest value) | $\frac{6.667}{0.669}$<br>= 9.96 | $\frac{9.300}{0.669}$<br>13.90 | $\frac{0.669}{0.669}$<br>1.00 | Round to Whole Number | 10 | 14 | 1 |
| Element                                          | C                               | H                              | O                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| %                                                | 80.0                            | 9.3                            | 10.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| No. of moles<br>(divide % by gfm)                | $\frac{80.0}{12}$<br>= 6.67     | $\frac{9.3}{1}$<br>= 9.300     | $\frac{10.7}{16}$<br>= 0.669                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| Mole ratio<br>(divide through by smallest value) | $\frac{6.667}{0.669}$<br>= 9.96 | $\frac{9.300}{0.669}$<br>13.90 | $\frac{0.669}{0.669}$<br>1.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| Round to Whole Number                            | 10                              | 14                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 25                                               | C                               | 58                             | <div><div><div>1<sup>st</sup> adjacent hydrogen</div><div>2<sup>nd</sup> adjacent hydrogen</div><div>3<sup>rd</sup> adjacent hydrogen</div></div><div></div></div> <div><div><br/>singlet<br/>no adjacent hydrogens</div><div><br/>Doublet<br/>1 adjacent hydrogen</div><div><br/>Triplet<br/>2 adjacent hydrogens</div><div><br/>quartet<br/>3 adjacent hydrogens</div><div><br/>quintet<br/>4 adjacent hydrogens</div></div>                                                                                                                                                                                                         |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 26                                               | A                               | 66                             | <div><input checked="" type="checkbox"/>A both are agonists as both bind with a receptor and stimulate a response</div> <div><input checked="" type="checkbox"/>B Buprenorphine is an agonist as it stimulates receptors and produces a response</div> <div><input checked="" type="checkbox"/>C Pramipexole is an agonist as it stimulates the nerves cells to make a response</div> <div><input checked="" type="checkbox"/>D Pramipexole is an agonist as it stimulates the nerves cells to make a response</div>                                                                                                                   |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 27                                               | D                               | 80                             | <div>1 minute = 6litres air ∴ 10minutes = 60litres air</div> <div>0.03ppm = 0.03mg per 1litre air</div> <div>1 litre of air ↔ 0.03mg hydrogen sulphide</div> <div>60 litres of air ↔ 0.03mg hydrogen sulphide × 60/1</div> <div>= 1.8mg</div>                                                                                                                                                                                                                                                                                                                                                                                          |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 28                                               | C                               | 87                             | <div><input checked="" type="checkbox"/>A Thin layer chromatography separates chemicals with different polarities/size</div> <div><input checked="" type="checkbox"/>B Heating under reflux allows volatile chemicals to react without reactant escape</div> <div><input checked="" type="checkbox"/>C Recrystallisation allows chemical to be extracted from impurities due to solubility in a second solvent</div> <div><input checked="" type="checkbox"/>D Distillation separates chemicals with different boiling points</div>                                                                                                    |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 29                                               | D                               | 70                             | <div><input checked="" type="checkbox"/>A water is less dense so should be the top layer</div> <div><input checked="" type="checkbox"/>B water is less dense so should be the top layer</div> <div><input checked="" type="checkbox"/>C <math>K = \frac{2}{8} = 0.25</math></div> <div><input checked="" type="checkbox"/>D <math>K = \frac{8}{2} = 4</math></div> <div><math>K = \frac{[X]_{\text{dichloromethane}}}{[X]_{\text{water}}}</math></div>                                                                                                                                                                                 |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |
| 30                                               | A                               | 52                             | <div><input checked="" type="checkbox"/>A Complexometric analysis is used to calculate the calcium ions in milk</div> <div><input checked="" type="checkbox"/>B Gravimetric analysis is used to calculate the chloride ions in sea water</div> <div><input checked="" type="checkbox"/>C Volumetric analysis is used to calculate ethanoic acid in vinegar</div> <div><input checked="" type="checkbox"/>D Volumetric analysis is used to calculate ethanol in wine</div>                                                                                                                                                              |         |   |   |   |   |      |     |      |                                   |                             |                            |                              |                                                  |                                 |                                |                               |                       |    |    |   |

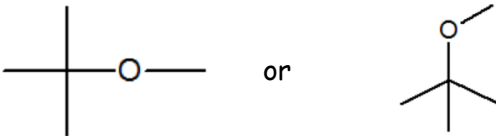
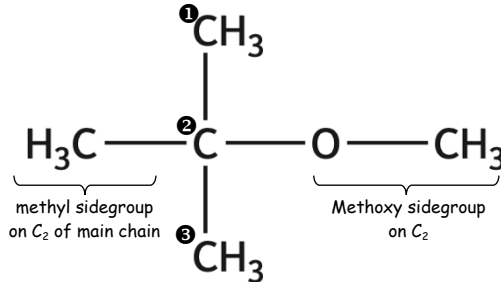
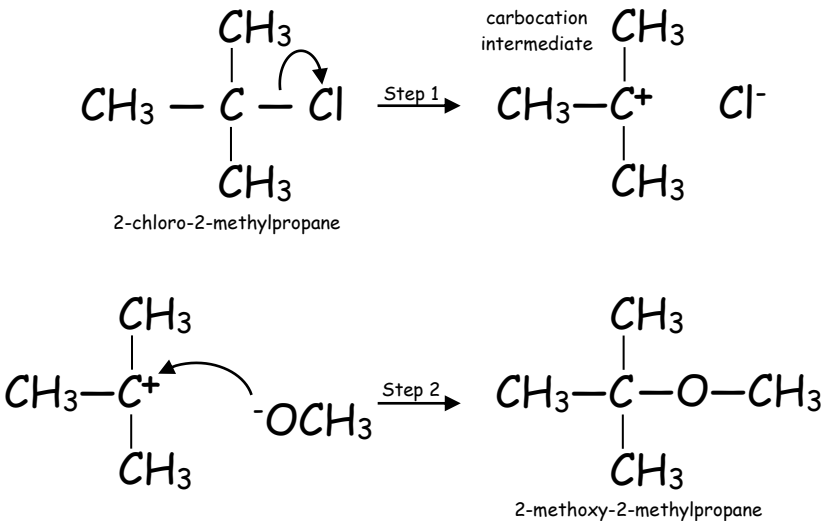
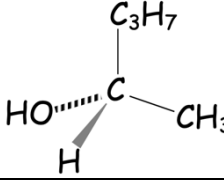
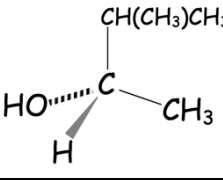
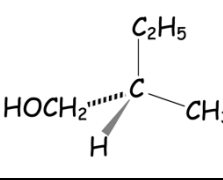
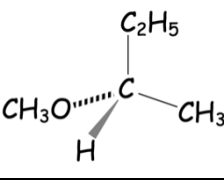
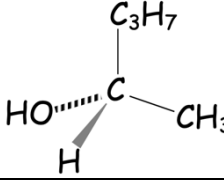
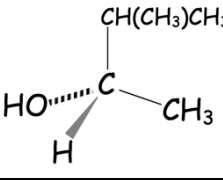
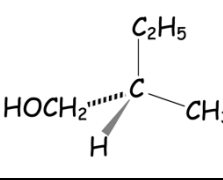
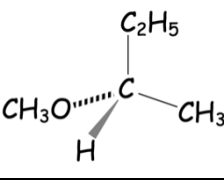
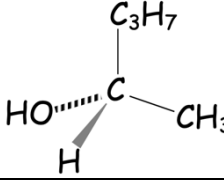
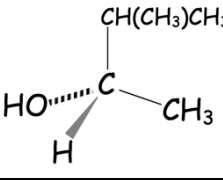
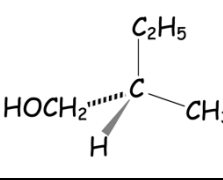
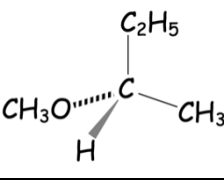
# 2019 Adv Higher Chemistry Marking Scheme

| Long Qu                                  | Answer                                                                                                    | Reasoning                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| 1a                                       | Any <u>one</u> of the 2p electrons circled                                                                | $n=2$<br>Electron in Shell 2                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $l=1$<br>Electron in p subshell                                  | $m_l = -1$<br>Electron in orbital -1<br><small>(could be any of 3 orbitals in diagram)</small>                                                         | $m_s = +\frac{1}{2}$<br>Electron has $+\frac{1}{2}$ spin<br><small>(could be either up or down in diagram)</small> |                                          |                                                    |                                                     |
| 1b                                       | 103                                                                                                       | $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ $= 127 - 298 \times \frac{79.4}{1000}$ $= 127 - 23.7$ $= 103.3 \text{ kJ mol}^{-1}$                                                                                                                                                                                                                                                                                                                                                |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 1c                                       | $3.20 \times 10^{-5}$                                                                                     | $\Delta G^\circ = -2.30 \times R \times T \times \log_{10}K$ $\log_{10}K = \frac{\Delta G^\circ}{-2.30 \times R \times T}$ $\log_{10}K = \frac{25.6}{-2.30 \times 8.31 \times 10^{-3} \times 298}$ $\log_{10}K = -4.49$ $K = 10^{-4.49} = 3.20 \times 10^{-5}$                                                                                                                                                                                                                         |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 2a                                       | Number of moles of reactant involved in the rate determining step                                         | Rate determining step is the slowest step in reaction mechanism. The number of moles of each reactant in rate determining step decides the order for each reactant <table><tr><td>0 moles of reactant in RDS<br/>Zero Order</td><td>1 mole of reactant in RDS<br/>1<sup>st</sup> Order</td><td>2 moles of reactant in RDS<br/>2<sup>nd</sup> Order</td></tr></table>                                                                                                                   |                                                                  |                                                                                                                                                        |                                                                                                                    | 0 moles of reactant in RDS<br>Zero Order | 1 mole of reactant in RDS<br>1 <sup>st</sup> Order | 2 moles of reactant in RDS<br>2 <sup>nd</sup> Order |
| 0 moles of reactant in RDS<br>Zero Order | 1 mole of reactant in RDS<br>1 <sup>st</sup> Order                                                        | 2 moles of reactant in RDS<br>2 <sup>nd</sup> Order                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 2b(i)                                    | 2 <sup>nd</sup> order                                                                                     | 1 mol of H <sub>2</sub> O <sub>2</sub> in rate determining step ∴ order [H <sub>2</sub> O <sub>2</sub> ] <sup>1</sup><br>1 mol of I <sup>-</sup> in rate determining step ∴ order [I <sup>-</sup> ] <sup>1</sup><br>Overall order = 1 + 1 = 2                                                                                                                                                                                                                                          |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 2b(ii)                                   | Rate = k [H <sub>2</sub> O <sub>2</sub> ] [I <sup>-</sup> ]                                               | Rate = k [H <sub>2</sub> O <sub>2</sub> ] <sup>1</sup> × [I <sup>-</sup> ] <sup>1</sup> = k [H <sub>2</sub> O <sub>2</sub> ] [I <sup>-</sup> ]                                                                                                                                                                                                                                                                                                                                         |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 2c                                       | $\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}_3\text{O}^+ \rightarrow \text{I}_2 + 4\text{H}_2\text{O}$ | <div>Step 1    H<sub>2</sub>O<sub>2</sub> + I<sup>-</sup> → <del>IO<sup>-</sup></del> + H<sub>2</sub>O</div> <div>Step 2    <del>IO<sup>-</sup></del> + H<sub>3</sub>O<sup>+</sup> → <del>HOI</del> + H<sub>2</sub>O</div> <div>Step 3    <del>HOI</del> + H<sub>3</sub>O<sup>+</sup> + I<sup>-</sup> → I<sub>2</sub> + 2H<sub>2</sub>O</div> <div>Overall    H<sub>2</sub>O<sub>2</sub> + 2I<sup>-</sup> + 2H<sub>3</sub>O<sup>+</sup> → I<sub>2</sub> + 4H<sub>2</sub>O</div>        |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 3a                                       | Answer to include:                                                                                        | 1 <sup>st</sup> Mark                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5cm <sup>3</sup>                                                 | As a $\frac{1}{10}$ dilution needs to take place, 5cm <sup>3</sup> is transferred to the 50cm <sup>3</sup> flask                                       |                                                                                                                    |                                          |                                                    |                                                     |
|                                          |                                                                                                           | 2 <sup>nd</sup> Mark                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Correct use of pipette<br>Making Up to line with deionised water | An accurate method of measuring 5cm <sup>3</sup> need to be used e.g. pipette<br>The volumetric/standard flask must be filled to the line for accuracy |                                                                                                                    |                                          |                                                    |                                                     |
| 3b(i)                                    | Deionised water                                                                                           | The control experiment has a cuvette containing deionised water and the absorbance of this is the setting for zero absorbance.                                                                                                                                                                                                                                                                                                                                                         |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 3b(ii)                                   | Unknown's Absorbance must not be outwith calibration range                                                | The calibration curve is only applicable to concentrations between the highest and lowest concentrations used in the calibration curve. Diluting a sample can bring a more concentrated unknown into the usable range of the calibration curve. It is unknown what happens to concentrations beyond the calibration range with any certainty.                                                                                                                                          |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |
| 3b(iii)                                  | 71%                                                                                                       | Absorbance (diluted sample) = 0.34 ∴ concentration of Cu <sup>2+</sup> = 0.032 mol l <sup>-1</sup><br>∴ concentration of Cu <sup>2+</sup> in original sample = 0.064 mol l <sup>-1</sup><br>no. of mol in 250cm <sup>3</sup> = volume × concentration = 0.25litres × 0.064 mol l <sup>-1</sup> = 0.016mol<br>mass = no. of mol × gfm = 0.016 × 63.5 = 1.016g<br>$\% \text{ mass} = \frac{\text{mass of Cu}}{\text{mass of screw}} \times 100 = \frac{1.016}{1.43} \times 100 = 71.0\%$ |                                                                  |                                                                                                                                                        |                                                                                                                    |                                          |                                                    |                                                     |

|                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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----------------|--|------------------------|--|-----------------------------------------|--|---------------------------------------|
| 4a(i)                                                                                                                                                                                                                                                                           | Proton or H <sup>+</sup> acceptor                                                                                                                                                                                                                                              | <table><tr><td>Acid</td><td>Donates H<sup>+</sup></td></tr><tr><td>Base</td><td>Accepts H<sup>+</sup></td></tr><tr><td>Conjugate Acid</td><td>Formed when Base accepts H<sup>+</sup></td></tr><tr><td>Conjugate Base</td><td>Formed when Acid loses H</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Acid                                                                                                                                                                                                                                                                            | Donates H <sup>+</sup>                                                                                                                                                                                  | Base                                                                                                                                                                                                                                                           | Accepts H <sup>+</sup>                | Conjugate Acid                | Formed when Base accepts H <sup>+</sup> | Conjugate Base                                                                                                                                                                                                                             | Formed when Acid loses H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                       |                                                      |                  |                                |                               |                                      |                              |                             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           |  |                        |  |                                         |  |                                       |
| Acid                                                                                                                                                                                                                                                                            | Donates H <sup>+</sup>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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           |  |                        |  |                                         |  |                                       |
| Base                                                                                                                                                                                                                                                                            | Accepts H <sup>+</sup>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Conjugate Acid                                                                                                                                                                                                                                                                  | Formed when Base accepts H <sup>+</sup>                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Conjugate Base                                                                                                                                                                                                                                                                  | Formed when Acid loses H                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 4a(ii)                                                                                                                                                                                                                                                                          | <table><tr><td>Acid</td><td>H<sub>2</sub>O<sub>2</sub></td></tr><tr><td>Conjugate Base</td><td>HO<sub>2</sub><sup>-</sup></td></tr></table> or <table><tr><td>Acid</td><td>H<sub>3</sub>O<sup>+</sup></td></tr><tr><td>Conjugate Base</td><td>H<sub>2</sub>O</td></tr></table> | Acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                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        | HO <sub>2</sub> <sup>-</sup>                                                                                                                                                                                                                                   | Acid                                  | H <sub>3</sub> O <sup>+</sup> | Conjugate Base                          | H <sub>2</sub> O                                                                                                                                                                                                                           | <table><tr><td>H<sub>2</sub>O<sub>2</sub></td><td>+</td><td>H<sub>2</sub>O</td><td>⇌</td><td>H<sub>3</sub>O<sup>+</sup></td><td>+</td><td>HO<sub>2</sub><sup>-</sup></td></tr><tr><td>Acid</td><td></td><td>Base</td><td></td><td>Conjugate Acid</td><td></td><td>Conjugate Base</td></tr><tr><td>Donates H<sup>+</sup></td><td></td><td>Accepts H<sup>+</sup></td><td></td><td>Formed when Base accepts H<sup>+</sup></td><td></td><td>Formed when Acid loses H<sup>+</sup></td></tr></table> or <table><tr><td>H<sub>3</sub>O<sup>+</sup></td><td>+</td><td>HO<sub>2</sub><sup>-</sup></td><td>⇌</td><td>H<sub>2</sub>O<sub>2</sub></td><td>+</td><td>H<sub>2</sub>O</td></tr><tr><td>Acid</td><td></td><td>Base</td><td></td><td>Conjugate Acid</td><td></td><td>Conjugate Base</td></tr><tr><td>Donates H<sup>+</sup></td><td></td><td>Accepts H<sup>+</sup></td><td></td><td>Formed when Base accepts H<sup>+</sup></td><td></td><td>Formed when Acid loses H<sup>+</sup></td></tr></table> | H <sub>2</sub> O <sub>2</sub>                                                                                                                                                         | +                                                    | H <sub>2</sub> O | ⇌                              | H <sub>3</sub> O <sup>+</sup> | +                                    | HO <sub>2</sub> <sup>-</sup> | Acid                                                                                                                      |                                                                            | Base                                                                                                                                                                                                                                       |        | Conjugate Acid |                          | Conjugate Base | Donates H <sup>+</sup>  |         | Accepts H <sup>+</sup>               |         | Formed when Base accepts H <sup>+</sup>                                                                                   |                                                                            | Formed when Acid loses H <sup>+</sup> | H <sub>3</sub> O <sup>+</sup> | + | HO <sub>2</sub> <sup>-</sup> | ⇌ | H <sub>2</sub> O <sub>2</sub> | + | H <sub>2</sub> O | Acid |  | Base |  | Conjugate Acid |  | Conjugate Base | Donates H <sup>+</sup> |  | Accepts H <sup>+</sup> |  | Formed when Base accepts H <sup>+</sup> |  | Formed when Acid loses H <sup>+</sup> |
| Acid                                                                                                                                                                                                                                                                            | H <sub>2</sub> O <sub>2</sub>                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Conjugate Base                                                                                                                                                                                                                                                                  | HO <sub>2</sub> <sup>-</sup>                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Acid                                                                                                                                                                                                                                                                            | H <sub>3</sub> O <sup>+</sup>                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| Conjugate Base                                                                                                                                                                                                                                                                  | H <sub>2</sub> O                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| H <sub>2</sub> O <sub>2</sub>                                                                                                                                                                                                                                                   | +                                                                                                                                                                                                                                                                              | H <sub>2</sub> O                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| Acid                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                | Base                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| Donates H <sup>+</sup>                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                | Accepts H <sup>+</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                              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| H <sub>3</sub> O <sup>+</sup>                                                                                                                                                                                                                                                   | +                                                                                                                                                                                                                                                                              | HO <sub>2</sub> <sup>-</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                        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| Acid                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                | Base                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| Donates H <sup>+</sup>                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                | Accepts H <sup>+</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                              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| 4b                                                                                                                                                                                                                                                                              | One answer from:                                                                                                                                                                                                                                                               | <table><tr><td>B(OH)<sub>3</sub> accepts a pair of non-bonding electrons and water donates a pair of non-bonding electrons</td><td>B(OH)<sub>3</sub> accepts a pair of non-bonding electrons from water.</td><td>Water donates a pair of non-bonding electrons to B(OH)<sub>3</sub></td></tr></table>                                                                                                                                                               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        | Water donates a pair of non-bonding electrons to B(OH) <sub>3</sub>                                                                                                                                                                                            |                                       |                               |                                         |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                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| B(OH) <sub>3</sub> accepts a pair of non-bonding electrons and water donates a pair of non-bonding electrons                                                                                                                                                                    | B(OH) <sub>3</sub> accepts a pair of non-bonding electrons from water.                                                                                                                                                                                                         | Water donates a pair of non-bonding electrons to B(OH) <sub>3</sub>                                                                                                                                                                                                                                                                                                                                                                                                 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| 4c(i)                                                                                                                                                                                                                                                                           | Increasing number of chlorine increases strength of acid                                                                                                                                                                                                                       | Increasing the number of chlorines increases the value of K <sub>a</sub> .<br>The higher the value of K <sub>a</sub> the stronger the acid<br>(NB The higher the value of pK <sub>a</sub> the weaker the acid)                                                                                                                                                                                                                                                      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| 4c(ii)<br>Part A                                                                                                                                                                                                                                                                | 0.08                                                                                                                                                                                                                                                                           | gfm CH <sub>2</sub> COOH = (2x12)+(3x1)+(2x16)+(1x35.5) = 24+3+32+35.5 = 94.5g<br>no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1.89}{94.5} = 0.02\text{mol}$<br>concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{0.02 \text{ mol}}{0.25 \text{ litres}} = 0.08 \text{ mol l}^{-1}$                                                                                                                                                          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| 4c(ii)<br>Part B                                                                                                                                                                                                                                                                | 2.0                                                                                                                                                                                                                                                                            | <table><tr><td>pH =</td><td><math>\frac{1}{2} \text{ pK}_a</math></td><td>-</td><td><math>\frac{1}{2} \log c</math></td></tr><tr><td>pH =</td><td><math>\frac{1}{2} \times -\log_{10} K_a</math></td><td>-</td><td><math>\frac{1}{2} \log_{10} c</math></td></tr><tr><td>pH =</td><td><math>-\frac{1}{2} \times \log_{10} (1.3 \times 10^{-3})</math></td><td>-</td><td><math>\frac{1}{2} \log_{10} (0.08)</math></td></tr><tr><td>pH =</td><td><math>-\frac{1}{2} \times (-2.89)</math></td><td>-</td><td><math>\frac{1}{2} \times (-1.10)</math></td></tr><tr><td>pH =</td><td>1.44</td><td>-</td><td>(-0.55)</td></tr><tr><td>pH =</td><td>1.99</td><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                   | pH =                                                                                                                                                                                                                                                                            | $\frac{1}{2} \text{ pK}_a$                                                                                                                                                                              | -                                                                                                                                                                                                                                                              | $\frac{1}{2} \log c$                  | pH =                          | $\frac{1}{2} \times -\log_{10} K_a$     | -                                                                                                                                                                                                                                          | $\frac{1}{2} \log_{10} c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | pH =                                                                                                                                                                                  | $-\frac{1}{2} \times \log_{10} (1.3 \times 10^{-3})$ | -                | $\frac{1}{2} \log_{10} (0.08)$ | pH =                          | $-\frac{1}{2} \times (-2.89)$        | -                            | $\frac{1}{2} \times (-1.10)$                                                                                              | pH =                                                                       | 1.44                                                                                                                                                                                                                                       | -      | (-0.55)        | pH =                     | 1.99           |                         |         |                                      |         |                                                                                                                           |                                                                            |                                       |                               |   |                              |   |                               |   |                  |      |  |      |  |                |  |                |                        |  |                        |  |                                         |  |                                       |
| pH =                                                                                                                                                                                                                                                                            | $\frac{1}{2} \text{ pK}_a$                                                                                                                                                                                                                                                     | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| pH =                                                                                                                                                                                                                                                                            | $\frac{1}{2} \times -\log_{10} K_a$                                                                                                                                                                                                                                            | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| pH =                                                                                                                                                                                                                                                                            | $-\frac{1}{2} \times \log_{10} (1.3 \times 10^{-3})$                                                                                                                                                                                                                           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| pH =                                                                                                                                                                                                                                                                            | $-\frac{1}{2} \times (-2.89)$                                                                                                                                                                                                                                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| pH =                                                                                                                                                                                                                                                                            | 1.44                                                                                                                                                                                                                                                                           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| pH =                                                                                                                                                                                                                                                                            | 1.99                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 4d                                                                                                                                                                                                                                                                              | Open Question to include:                                                                                                                                                                                                                                                      | <table><tr><td><b>3 mark answer</b><br/>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td><b>2 mark answer</b><br/>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td><b>1 mark answer</b><br/>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>                                                                                                                                                                                                               | <b>3 mark answer</b><br>Demonstrates a <u>good understanding</u> of the chemistry involved. 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| 5a                                                                                                                                                                                                                                                                              | Heat or high temperature                                                                                                                                                                                                                                                       | Electrons become excited in sodium by absorbing heat energy which promotes an electron to a higher energy level. Light of specific wavelength is emitted corresponding to the exact energy difference between the upper energy level the electron drops down from to the lower energy level the electron goes in to.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                |                                       |                               |                                         |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                       |                                                      |                  |                                |                               |                                      |                              |                                                                                                                           |                                                                            |                                                                                                                                                                                                                                            |        |                |                          |                |                         |         |                                      |         |                                                                                                                           |                                                                            |                                       |                               |   |                              |   |                               |   |                  |      |  |      |  |                |  |                |                        |  |                        |  |                                         |  |                                       |
| 5b(i)                                                                                                                                                                                                                                                                           | Hexaaquazinc(II)                                                                                                                                                                                                                                                               | <p>Hexaaquazinc(II) = [Zn(H<sub>2</sub>O)<sub>6</sub>]<sup>2+</sup></p> <table><tr><td>no. of ligands</td><td>H<sub>2</sub>O ligand</td><td>metal name</td><td>Charge on metal ion</td></tr></table> <table><tr><td>Neutral ligands include:</td><td>Negative Ligands include:</td><td>Central Ion:</td><td>Charge:</td></tr><tr><td><table><tr><td>Ligand</td><td>Name</td></tr><tr><td>OH<sub>2</sub></td><td>aqua</td></tr><tr><td>NH<sub>3</sub></td><td>ammine</td></tr><tr><td>CO</td><td>carbonyl</td></tr></table></td><td><table><tr><td>Ligand</td><td>Name</td></tr><tr><td>Chloride Cl<sup>-</sup></td><td>chlorido</td></tr><tr><td>Cyanide CN<sup>-</sup></td><td>cyanido</td></tr><tr><td>Nitrite NO<sub>2</sub><sup>-</sup></td><td>nitrito</td></tr></table></td><td>Positive Complex:<br/>metals keep their name<br/>Negative Complex:<br/>Metals end in ATE<br/>e.g. Cuprate, Ferrate, Cobaltate</td><td>Charge of central ion is converted into roman numerals and put in brackets</td></tr></table> | no. of ligands                                                                                                                                                                                                                                                                  | H <sub>2</sub> O ligand                                                                                                                                                                                 | metal name                                                                                                                                                                                                                                                     | Charge on metal ion                   | Neutral ligands include:      | Negative Ligands include:               | Central Ion:                                                                                                                                                                                                                               | Charge:                                                                                                                                                                                                                   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| OH <sub>2</sub>                | aqua                          | NH <sub>3</sub>                      | ammine                       | CO                                                                                                                        | carbonyl                                                                   | <table><tr><td>Ligand</td><td>Name</td></tr><tr><td>Chloride Cl<sup>-</sup></td><td>chlorido</td></tr><tr><td>Cyanide CN<sup>-</sup></td><td>cyanido</td></tr><tr><td>Nitrite NO<sub>2</sub><sup>-</sup></td><td>nitrito</td></tr></table> | Ligand | Name           | Chloride Cl <sup>-</sup> | chlorido       | Cyanide CN <sup>-</sup> | cyanido | Nitrite NO <sub>2</sub> <sup>-</sup> | nitrito | Positive Complex:<br>metals keep their name<br>Negative Complex:<br>Metals end in ATE<br>e.g. Cuprate, Ferrate, Cobaltate | Charge of central ion is converted into roman numerals and put in brackets |                                       |                               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| no. of ligands                                                                                                                                                                                                                                                                  | H <sub>2</sub> O ligand                                                                                                                                                                                                                                                        | metal name                                                                                                                                                                                                                                                                                                                                                                                                                                                          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        | NH <sub>3</sub>                                                                                                                                                                                                                                                | ammine                                | CO                            | carbonyl                                | <table><tr><td>Ligand</td><td>Name</td></tr><tr><td>Chloride Cl<sup>-</sup></td><td>chlorido</td></tr><tr><td>Cyanide CN<sup>-</sup></td><td>cyanido</td></tr><tr><td>Nitrite NO<sub>2</sub><sup>-</sup></td><td>nitrito</td></tr></table> | Ligand                                                                                                                                                                                                                                                                                                                                                                                         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| Cyanide CN <sup>-</sup>                                                                                                                                                                                                                                                         | cyanido                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 5b(ii)                                                                                                                                                                                                                                                                          | Answer to include:                                                                                                                                                                                                                                                             | <table><tr><td>1<sup>st</sup> Mark<br/>Full or complete d subshell or 10 d electrons or 3d<sup>10</sup> or no d-d transitions possible</td><td>2<sup>nd</sup> Mark<br/>It does not absorb visible light or<br/>It only absorbs ultraviolet/UV light</td></tr></table>                                                                                                                                                                                               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| 1 <sup>st</sup> Mark<br>Full or complete d subshell or 10 d electrons or 3d <sup>10</sup> or no d-d transitions possible                                                                                                                                                        | 2 <sup>nd</sup> Mark<br>It does not absorb visible light or<br>It only absorbs ultraviolet/UV light                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                 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|-----------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------|-------------------------------------|------|------------------------------------|-----------------------------------|----------------------------------|------------------------------|---|-------|-------|---|---------|---|---------|-------|---|-------|--|--|
| 5c(i)                             | $2.04 \times 10^{-16} \text{ J}$                      | $E = h \times f$<br>$E = 6.63 \times 10^{-34} \text{ J s} \times 3.08 \times 10^{17} \text{ s}^{-1}$<br>$E = 2.04 \times 10^{-16} \text{ J}$<br><small>NB: The question does not require you to multiple by L (<math>6.02 \times 10^{23}</math>) as the units required are J not <math>\text{J mol}^{-1}</math> or <math>\text{kJ mol}^{-1}</math></small>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 5c(ii)                            | 19                                                    | $1 \text{ J} \longleftrightarrow 6.24 \times 10^{18} \text{ eV}$<br>$2.04 \times 10^{-16} \text{ J} \longleftrightarrow 6.24 \times 10^{18} \text{ eV} \times 2.04 \times 10^{-16} /_1$<br>$= 1273 \text{ eV}$<br><table><tr><td>Binding Energy</td><td>=</td><td>Energy of electromagnetic radiation</td><td>-</td><td>Kinetic Energy of electron emitted</td></tr><tr><td><math>E_b</math></td><td>=</td><td><math>E</math></td><td>-</td><td><math>E_k</math></td></tr><tr><td><math>E_b</math></td><td>=</td><td>1273 eV</td><td>-</td><td>1254 eV</td></tr><tr><td><math>E_b</math></td><td>=</td><td>19 eV</td><td></td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Binding Energy               | =                                  | Energy of electromagnetic radiation | -    | Kinetic Energy of electron emitted | $E_b$                             | =                                | $E$                          | - | $E_k$ | $E_b$ | = | 1273 eV | - | 1254 eV | $E_b$ | = | 19 eV |  |  |
| Binding Energy                    | =                                                     | Energy of electromagnetic radiation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                            | Kinetic Energy of electron emitted |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| $E_b$                             | =                                                     | $E$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                            | $E_k$                              |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| $E_b$                             | =                                                     | 1273 eV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -                            | 1254 eV                            |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| $E_b$                             | =                                                     | 19 eV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 6a                                | 6.4                                                   | Total Dichromate in flask:<br>no. of mol = volume $\times$ concentration = $0.025 \text{ litres} \times 0.010 \text{ mol l}^{-1} = 2.5 \times 10^{-4} \text{ mol}$<br>Dichromate left at end of reaction:<br>no. of mol = $1.65 \times 10^{-4} \text{ mol}$ (in question)<br>Dichromate which reacted with ethanol:<br>no. of mol = $2.5 \times 10^{-4} \text{ mol} - 1.65 \times 10^{-4} \text{ mol} = 8.5 \times 10^{-5} \text{ mol}$<br>$3\text{C}_2\text{H}_5\text{OH} + 2\text{Cr}_2\text{O}_7^{2-} + 16\text{H}^+ \longrightarrow 3\text{CH}_3\text{COOH} + 4\text{Cr}^{3+} + 11\text{H}_2\text{O}$ <table><tr><td>3mol</td><td>2mol</td><td></td><td></td><td></td></tr><tr><td><math>1.28 \times 10^{-4} \text{ mol}</math></td><td><math>8.5 \times 10^{-5} \text{ mol}</math></td><td></td><td></td><td></td></tr></table><br>$1 \text{ cm}^3 \text{ diluted vodka} \longleftrightarrow 1.28 \times 10^{-4} \text{ mol}$<br>$1000 \text{ cm}^3 \text{ diluted vodka} \longleftrightarrow 1.28 \times 10^{-4} \text{ mol} \times 1000 /_1$<br>$= 0.128 \text{ mol}$<br>As $1000 \text{ cm}^3$ diluted vodka was made from $20.0 \text{ cm}^3$ of vodka<br>$\therefore 20 \text{ cm}^3 \text{ undiluted vodka} \longleftrightarrow 0.128 \text{ mol}$<br>$\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.128 \text{ mol}}{0.020 \text{ litres}} = 6.4 \text{ mol l}^{-1}$ | 3mol                         | 2mol                               |                                     |      |                                    | $1.28 \times 10^{-4} \text{ mol}$ | $8.5 \times 10^{-5} \text{ mol}$ |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 3mol                              | 2mol                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| $1.28 \times 10^{-4} \text{ mol}$ | $8.5 \times 10^{-5} \text{ mol}$                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 6b                                | To ensure all ethanol is reacted                      | This experiment is a back titration. An excess of acidified potassium dichromate is added to ethanol solution and all the ethanol will react with acidified potassium dichromate solution and the left over acidified potassium dichromate can be determined by volumetric analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 6c                                | Impurities in vodka reacting with dichromate          | Other chemicals in the vodka may be able to react with acidified dichromate (a powerful oxidising agent).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 6d                                | Use standard solution of ethanol                      | A standard solution of ethanol could be used as a control to measure the experimentally-determined value of this ethanol solution against the known value of the standard solution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 7a                                | Under suction/vacuum                                  | A Buchner funnel and suction pump is used to draw the filtrate through the filter paper in the Buchner funnel quicker than gravity alone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 7b(i)                             | End on overlap of two atomic orbitals                 | A sigma bond is formed by the end on overlap of two orbitals.<br>A pi bond is formed by side overlap of two unhybridized p orbitals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                              |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| 7b(ii)                            | Mixing an s atomic orbital with two p atomic orbitals | <table><tr><td>Type of Hybridisation</td><td><math>sp^3</math></td><td><math>sp^2</math></td><td><math>sp</math></td></tr><tr><td>Description</td><td>One s and three p orbitals mix</td><td>One s and two p orbitals mix</td><td>One s and one p orbitals mix</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Type of Hybridisation        | $sp^3$                             | $sp^2$                              | $sp$ | Description                        | One s and three p orbitals mix    | One s and two p orbitals mix     | One s and one p orbitals mix |   |       |       |   |         |   |         |       |   |       |  |  |
| Type of Hybridisation             | $sp^3$                                                | $sp^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $sp$                         |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |
| Description                       | One s and three p orbitals mix                        | One s and two p orbitals mix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | One s and one p orbitals mix |                                    |                                     |      |                                    |                                   |                                  |                              |   |       |       |   |         |   |         |       |   |       |  |  |

| 7c                                                                                                                                                                                                                                                      | Answer to include:                                                                                                                                                              | <table><tr><th>1<sup>st</sup> Mark</th><th>2<sup>nd</sup> Mark</th></tr><tr><td>Electrons promoted/move from HOMO to LUMO</td><td>Blue/green light is absorbed or<br/>Complementary colour to red absorbed</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 <sup>st</sup> Mark                                              | 2 <sup>nd</sup> Mark                                      | Electrons promoted/move from HOMO to LUMO                                                               | Blue/green light is absorbed or<br>Complementary colour to red absorbed                                                                                                                                                                                 |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 <sup>st</sup> Mark                                                                                                                                                                                                                                    | 2 <sup>nd</sup> Mark                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| Electrons promoted/move from HOMO to LUMO                                                                                                                                                                                                               | Blue/green light is absorbed or<br>Complementary colour to red absorbed                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 7d(i)                                                                                                                                                                                                                                                   | One answer from:                                                                                                                                                                | <table><tr><td>Similar polarities or both polar or similar intermolecular forces</td><td>Forms hydrogen bonds with alizarin</td><td>Does not react with alizarin</td><td>Volatile or low boiling point or evaporates easily</td><td>Dissolves alizarin but not other substances (in the root)</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Similar polarities or both polar or similar intermolecular forces | Forms hydrogen bonds with alizarin                        | Does not react with alizarin                                                                            | Volatile or low boiling point or evaporates easily                                                                                                                                                                                                      | Dissolves alizarin but not other substances (in the root)                                                                                                                       |                                                                                                                                                                                                                                        |
| Similar polarities or both polar or similar intermolecular forces                                                                                                                                                                                       | Forms hydrogen bonds with alizarin                                                                                                                                              | Does not react with alizarin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Volatile or low boiling point or evaporates easily                | Dissolves alizarin but not other substances (in the root) |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 7d(ii)<br>Part A                                                                                                                                                                                                                                        | Answer to include:                                                                                                                                                              | <table><tr><th>1<sup>st</sup> Mark</th><th>2<sup>nd</sup> Mark</th></tr><tr><td>IR makes bonds vibrate/bend/stretch</td><td>Different bonds/functional groups absorb at different wavenumber/wavelength/frequency/energy</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 <sup>st</sup> Mark                                              | 2 <sup>nd</sup> Mark                                      | IR makes bonds vibrate/bend/stretch                                                                     | Different bonds/functional groups absorb at different wavenumber/wavelength/frequency/energy                                                                                                                                                            |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 1 <sup>st</sup> Mark                                                                                                                                                                                                                                    | 2 <sup>nd</sup> Mark                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| IR makes bonds vibrate/bend/stretch                                                                                                                                                                                                                     | Different bonds/functional groups absorb at different wavenumber/wavelength/frequency/energy                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 7d(ii)<br>Part B                                                                                                                                                                                                                                        | One or both -OH groups circled                                                                                                                                                  | <p>From page 14 of data booklet:</p> <table><tr><th>Wave Number Range/cm<sup>-1</sup></th><th>Type of Compound</th><th>Infra-red Absorption due to</th></tr><tr><td>3570-3200</td><td>Alcohols and Phenols</td><td>Hydrogen bonded O - H stretch</td></tr></table> <p>The -OH hydroxyl group can be described as phenols as they are attached to a benzene ring.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       | Wave Number Range/cm <sup>-1</sup>                                | Type of Compound                                          | Infra-red Absorption due to                                                                             | 3570-3200                                                                                                                                                                                                                                               | Alcohols and Phenols                                                                                                                                                            | Hydrogen bonded O - H stretch                                                                                                                                                                                                          |
| Wave Number Range/cm <sup>-1</sup>                                                                                                                                                                                                                      | Type of Compound                                                                                                                                                                | Infra-red Absorption due to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 3570-3200                                                                                                                                                                                                                                               | Alcohols and Phenols                                                                                                                                                            | Hydrogen bonded O - H stretch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 7d(ii)<br>Part C (I)                                                                                                                                                                                                                                    | 2.96x10 <sup>-6</sup> m                                                                                                                                                         | $\text{Wavelength} = \frac{1}{\text{Wavenumber}} = \frac{1}{3395\text{cm}^{-1}} = 2.96 \times 10^{-4} \text{ cm} = 2.96 \times 10^{-6} \text{ m}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 7d(ii)<br>Part C (II)                                                                                                                                                                                                                                   | 40.65kJ mol <sup>-1</sup>                                                                                                                                                       | $E = \frac{L \times h \times c}{\lambda} = \frac{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{2.96 \times 10^{-6} \text{ m}}$ $= 40651 \text{ J mol}^{-1}$ $= 40.65 \text{ kJ mol}^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 8                                                                                                                                                                                                                                                       | Open Question to include:                                                                                                                                                       | <table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table> | 3 mark answer                                                     | 2 mark answer                                             | 1 mark answer                                                                                           | Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem. | Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood. | Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. |
| 3 mark answer                                                                                                                                                                                                                                           | 2 mark answer                                                                                                                                                                   | 1 mark answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem. | Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood. | Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 9a(i)                                                                                                                                                                                                                                                   | Either circle on diagram:<br>                                                                | A chiral carbon has four different chemical groups attached to a central carbon. Both circled carbons have four bond but due to being skeletal formula diagrams the 4th bond (a C-H bond) being the 4 <sup>th</sup> different group attached to the central carbon is not drawn as it is a C-H bond.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 9a(ii)                                                                                                                                                                                                                                                  | Non-superimposable mirror images                                                                                                                                                | <table><tr><th>Non-superimposable Mirror Images</th><th>Non-superimposable Mirror Images</th></tr><tr><td>Images shown are direct mirror images of each other and are therefore are optical isomers (enantiomers)</td><td>The following are optical isomers as Groups W and X are in the same position but Groups Y and Z are in different position.</td></tr><tr><td></td><td></td></tr></table>                                                                                                                                                                                                                                 | Non-superimposable Mirror Images                                  | Non-superimposable Mirror Images                          | Images shown are direct mirror images of each other and are therefore are optical isomers (enantiomers) | The following are optical isomers as Groups W and X are in the same position but Groups Y and Z are in different position.                                                                                                                              |                                                                                             |                                                                                                                                                   |
| Non-superimposable Mirror Images                                                                                                                                                                                                                        | Non-superimposable Mirror Images                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| Images shown are direct mirror images of each other and are therefore are optical isomers (enantiomers)                                                                                                                                                 | The following are optical isomers as Groups W and X are in the same position but Groups Y and Z are in different position.                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
|                                                                                                                                                                     |                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
| 9b(i)                                                                                                                                                                                                                                                   | (nucleophilic) substitution                                                                                                                                                     | <p>The lone pair on the pair of the N in NH<sub>3</sub> is nucleophilically attracted to the δ<sup>+</sup> on the C of the polar C-Br bond.</p> <p>The NH<sub>3</sub> is added onto the carbon as the Br atom leaves as a Br<sup>-</sup> ion.</p> <p>An H in the -NH<sub>3</sub><sup>+</sup> group then leaves as an H<sup>+</sup> ion leaving a -NH<sub>2</sub> amine group attached to the carbon the Br previously was attached to.</p>                                                                                                                                                                                                                                                                                                                                                                 |                                                                   |                                                           |                                                                                                         |                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                                        |

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|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 9b(ii)                                                                                                                | 7.58g                                                                                                                                                           | <div>1-phenylpropanone <math>\longrightarrow</math> [intermediate] <math>\longrightarrow</math> cathinone</div> <div>1mol<br/>134g<br/>9.50g</div> <div>1mol<br/>149g<br/><math>149g \times 9.50/134</math><br/><math>= 10.56g</math></div> <div><math>\%Yield = \frac{Actual}{Theoretical} \times 100 \therefore Actual = \frac{\% Yield \times Theoretical}{100} = \frac{71.8 \times 10.56}{100} = 7.58g</math></div>                                                                                                                                             |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10a                                                                                                                   | Ether                                                                                                                                                           | Ethers have a carbon-oxygen-carbon group as their functional group.<br>Be careful not to confuse a C-O-C group with an ester group which also has a carbonyl C=O group in one end.                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10b(i)                                                                                                                | One from:                                                                                                                                                       | <div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10b(ii)                                                                                                               | 2-methoxy-2-methylpropane                                                                                                                                       | <div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10c(i)                                                                                                                | methanol                                                                                                                                                        | Group 1 metals react with alcohols to produce alkoxides and hydrogen:<br><div>sodium + methanol <math>\longrightarrow</math> sodium methoxide + hydrogen<br/><math>2Na + 2CH_3OH \longrightarrow 2Na^+CH_3O^- + H_2</math></div>                                                                                                                                                                                                                                                                                                                                    |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10c(ii)                                                                                                               | <div><div><u>1<sup>st</sup> Mark</u><br/>Correct Carbocation intermediate</div><div><u>2<sup>nd</sup> Mark</u><br/>Correct use of both curly arrows</div></div> | <div><div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10c(iii)                                                                                                              | One answer from:                                                                                                                                                | <table><tr><td>Formation of stable (tertiary) carbocation</td><td>Steric hindrance of three methyl -CH<sub>3</sub> groups prevents nucleophilic attack</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                            | Formation of stable (tertiary) carbocation                                                                                  | Steric hindrance of three methyl -CH <sub>3</sub> groups prevents nucleophilic attack                                         |                                                                                                                                |                                                                                                                             |
| Formation of stable (tertiary) carbocation                                                                            | Steric hindrance of three methyl -CH <sub>3</sub> groups prevents nucleophilic attack                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10d                                                                                                                   | Any one from:                                                                                                                                                   | <table><tr><td><div></div><div>pentan-2-ol</div></td><td><div></div><div>3-methylbutan-2-ol</div></td><td><div></div><div>2-methylbutan-1-ol</div></td><td><div></div><div>2-methoxybutane</div></td></tr></table> | <div></div> <div>pentan-2-ol</div>       | <div></div> <div>3-methylbutan-2-ol</div> | <div></div> <div>2-methylbutan-1-ol</div> | <div></div> <div>2-methoxybutane</div> |
| <div></div> <div>pentan-2-ol</div> | <div></div> <div>3-methylbutan-2-ol</div>                                   | <div></div> <div>2-methylbutan-1-ol</div>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div></div> <div>2-methoxybutane</div> |                                                                                                                               |                                                                                                                                |                                                                                                                             |
| 10e                                                                                                                   | One line at 1.5-0.9 with relative intensity 9                                                                                                                   | <div>The peak drawn is 3.6-3.7 and a relative intensity of 3<br/><math>\therefore</math> peak caused -O-CH<sub>3</sub> group (3xH = relative intensity 3)</div> <div>Other Peak must contain tertiary -C(CH<sub>3</sub>)<sub>3</sub> group<br/><math>\therefore</math> Line at chemical shift 1.5-0.9 for R<sub>3</sub>C- from table<br/><math>\therefore</math> Relative intensity of 9 as there are 9 H atoms in group.</div>                                                                                                                                     |                                                                                                                             |                                                                                                                               |                                                                                                                                |                                                                                                                             |

2019 Advanced Higher Chemistry Section 1 Multiple Choice Analysis

| Item | Key      | A         | B         | C         | D         | Missing | Invalid |
|------|----------|-----------|-----------|-----------|-----------|---------|---------|
| 1    | <b>B</b> | 10        | <b>76</b> | 4         | 10        | 0       | 0       |
| 2    | <b>D</b> | 6         | 4         | 5         | <b>85</b> | 0       | 0       |
| 3    | <b>C</b> | 10        | 27        | <b>37</b> | 25        | 1       | 0       |
| 4    | <b>A</b> | <b>90</b> | 5         | 5         | 0         | 0       | 0       |
| 5    | <b>D</b> | 3         | 5         | 17        | <b>75</b> | 0       | 0       |
| 6    | <b>D</b> | 11        | 13        | 24        | <b>51</b> | 0       | 0       |
| 7    | <b>B</b> | 22        | <b>71</b> | 2         | 5         | 0       | 0       |
| 8    | <b>D</b> | 3         | 6         | 31        | <b>60</b> | 0       | 0       |
| 9    | <b>A</b> | <b>72</b> | 17        | 3         | 8         | 0       | 0       |
| 10   | <b>C</b> | 1         | 1         | <b>72</b> | 26        | 0       | 0       |
| 11   | <b>C</b> | 1         | 5         | <b>89</b> | 4         | 0       | 0       |
| 12   | <b>B</b> | 31        | <b>39</b> | 12        | 17        | 0       | 0       |
| 13   | <b>A</b> | <b>86</b> | 1         | 8         | 5         | 0       | 0       |
| 14   | <b>B</b> | 12        | <b>75</b> | 9         | 4         | 0       | 0       |
| 15   | <b>C</b> | 5         | 16        | <b>78</b> | 1         | 0       | 0       |
| 16   | <b>A</b> | <b>89</b> | 6         | 3         | 1         | 0       | 0       |
| 17   | <b>C</b> | 3         | 31        | <b>49</b> | 16        | 0       | 0       |
| 18   | <b>C</b> | 15        | 1         | <b>80</b> | 3         | 0       | 0       |
| 19   | <b>D</b> | 7         | 20        | 14        | <b>59</b> | 0       | 0       |
| 20   | <b>B</b> | 8         | <b>78</b> | 10        | 4         | 0       | 0       |
| 21   | <b>C</b> | 8         | 24        | <b>56</b> | 12        | 0       | 0       |
| 22   | <b>B</b> | 8         | <b>63</b> | 16        | 12        | 0       | 0       |
| 23   | <b>B</b> | 11        | <b>64</b> | 14        | 12        | 1       | 0       |
| 24   | <b>A</b> | <b>90</b> | 3         | 4         | 3         | 1       | 0       |
| 25   | <b>C</b> | 14        | 18        | <b>58</b> | 9         | 0       | 0       |
| 26   | <b>A</b> | <b>66</b> | 31        | 2         | 1         | 0       | 0       |
| 27   | <b>D</b> | 3         | 10        | 6         | <b>80</b> | 1       | 0       |
| 28   | <b>C</b> | 2         | 6         | <b>87</b> | 4         | 0       | 0       |
| 29   | <b>D</b> | 2         | 14        | 14        | <b>70</b> | 0       | 0       |
| 30   | <b>A</b> | <b>52</b> | 26        | 11        | 11        | 0       | 0       |